



Guided Work Series N°2

Exercise 01 (BCD code)

1. Convert each of the following numbers from their given base to BCD, showing all necessary steps of conversion: $(259)_{10}$, $(101101)_2$, $(1101011.101)_2$, $(745)_8$, $(3A7)_{16}$
2. Perform the following BCD additions, showing all steps (i.e. binary addition and correction (adding 0110 when needed)).
 $(5_{BCD} + 6_{BCD})$; $(54_{BCD} + 29_{BCD})$; $(134_{BCD} + 35_{BCD})$

Exercise 02 (BCD code)

1. Convert the numbers $A = (1447)_{10}$ and $B = (2025)_{10}$ to BCD and perform the operation $(A + B)$ in BCD. (Verify that the result is identical to the same operation performed in decimal).

Exercise 03 (XS-3 code)

1. Convert each of the following numbers to its Excess-3 (XS-3) representation
 $(47)_{10}$, $(101101)_2$, $(0100\ 1001)_{BCD}$, $(1001\ 0101\ 0111)_{BCD}$
2. Perform the following XS3 additions : $(5)_{10} + (3)_{10}$, $(123)_{10} + (259)_{10}$

Exercise 04 (GRAY code)

1. Convert the following **binary numbers** into their **Gray code** equivalents.
 $(101101)_2$, $(1100101)_2$, $(1000110110)_2$
2. Convert the following Gray code numbers into their Binary equivalents.
 $(110111)_{GRAY}$, $(1010101)_{GRAY}$, $(1110010110)_{GRAY}$

Exercise 05

1. What is the size, in bytes, of a text containing n characters encoded in the following formats: UTF-8; UTF-16 and UTF-32?
2. What is the ASCII code of: **Good Morning**
3. What is the ASCII code of: **Hexadecimal System**
4. Transcode the following sequence using ASCII code : **4D 61 74 68 65 6D 61 74 69 63 73 20 69 73 20 74 68 65 20 6C 61 6E 67 75 61 67 65 20 6F 66 20 6C 6F 67 69 63 20 61 6E 64 20 70 72 65 63 69 73 69 6F 6E**

Exercise 06 (SAV Representation)

Represent the following given integers in 9-bit SAV format:
+8; +39; +77; +199; -8; -39; -77; -199;

Exercise 07

Encode the integers 63, 1, 0, -1, -2, -63, and -64 using 7 bits with the following representations:

1. Sign-and-Magnitude (SAV)
2. One's complement,
3. Two's complement.

Exercise 08

1. What are the equivalent decimal numbers of the following 16-bit (1C) representations?
a. 1110010101010100 b. 0000000010101001
2. What are the equivalent decimal numbers of the following 16-bit (2C) representations?
a. 1111111100100110 b. 0000000001111010 c. 1111111110001101
3. Perform the following operations in (1C) representation (determine the number of bits needed to perform each operation)
 $(7-13)_{10}$; $(-5-9)_{10}$; $(13+56)_{10}$
4. Perform the following operations in (2C) representation (determine the number of bits needed to perform each operation)
 $(-4+11)_{10}$; $(9-6)_{10}$; $(-10-29)_{10}$

Exercise 09 (fixed-point representation)

- Express the following numbers in 12-bit SAV fixed-point format, using 3 fractional bits
 $(4.125)_{10}$; $(-99.8125)_{10}$; (-0.75)
- Determine the real numbers represented by the following SAV fixed-point numbers. Each value is a 16-bit word, including a sign bit and 6 fractional bits.
 - 0000101010011000
 - 1000011100100000
 - 1000000111111100

Exercise 10 (floating-point representation)

- Express the following numbers in 16-bit floating-point format, using 6 bits for the exponent.
 $(-3.125)_{10}$; $(76.375)_{10}$; $(140.5)_{10}$
- Determine the real numbers represented by the following 16-bit floating-point numbers, which include a sign bit and 6 bits for exponent.
 - 0100001101000000
 - 1100011111110000
 - 1011110110000000

Exercise 11 (floating-point representation IEEE 754 single-precision)

- Express the following numbers using the IEEE 754 single-precision floating-point format.
 - $(-128,25)_{10}$,
 - $(+52,34)_8$,
 - $(-0,375)_{10}$
- Determine the decimal values of the IEEE 754 single-precision floating-point numbers.
 - $(1100\ 0011\ 1101\ 1010\ 0000\ 0000\ 0000\ 0000)_2$
 - $(0100\ 0001\ 1101\ 1000\ 0000\ 0000\ 0000\ 0000)_2$
 - $(438DD000)_H$

Exercise 12 (floating-point representation IEEE 754 Double-precision)

- Express the following numbers using the IEEE 754 double-precision floating-point format.
 - $(-2025)_{10}$,
 - $(-7,625)_{10}$,
 - $(3C)_H$

2. Determine the decimal values of the IEEE 754 double-precision floating-point numbers.

- $a = (\text{C033 8000 0000 0000})_{\text{H}}$
- $b = (\text{4059 4600 0000 0000})_{\text{H}}$
- $c = (\text{BFE8 0000 0000 0000})_{\text{H}}$

Exercise 13 (floating-point representation IEEE 754: Arithmetic operations)

Let **p**, **q**, **r**, and **s** be 32-bit IEEE 754 single-precision floating-point numbers represented in **hexadecimal**:

- $p = (\text{42160000})_{16}$
- $q = (\text{C0A00000})_{16}$
- $r = (\text{40F00000})_{16}$
- $s = (\text{BF800000})_{16}$

Required:

Compute the following floating-point operations:

1. $p \div q$
2. $r \times s$
3. $p - r$
4. $q + s$